

SIEMENS

MAGNETOM

MR

Troubleshooting Guide

Gradient Error Messages

For all Tim Systems
with Software Version syngo MR B17 (VB17A)

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Print No.: MR00-000.840.16.01.02
Replaces: n.a.

English
Doc. Gen. Date: 03.09

History - Changes to previous version

Date	Author	Version	Changes
03.09	CS SD MR	1.0	Initial version

Gradient

Message class	Id	Source	Message Text	Explanation	Action	Information
S/I/W/E						
Error	17	GXXmsg.mc	Gradient Power Amplifier error: Refer to other messages for details. If the problem still persists, please call Siemens Service.	None.	Do actions according to error messages in the eventlog.	None.
Error	18	GXXmsg.mc	Gradient Power Amplifier error: Power supply error (+5V). Please call Siemens Service	Plus 5V supply voltage is out of tolerance.	1.a) Measure (5.05V at D70 X63 to X8)/adjust power supply slot 2 or replace the power supply. 1.b) Check the fuse F1 on board D70 1.c) If error occurs again, please check all PCB's in the GSSU 2.) Error of GSSU 5VDC power supply; circuit breaker F2 in the GPA is switched off; low line voltage. 3.) Check for monitoring error.	None.
Error	19	GXXmsg.mc	Gradient Power Amplifier error: Power supply error (+/-15V). Please call Siemens Service	Plus or Minus 15V supply voltage is out of tolerance.	1.a) Measure (15.05V at D70 X61 to X8, -15.05V at D70 X65 to X8)/adjust (power supply slot 3) or replace the power supply. 1.b) If error occurs again, please check all PCB's in the GSSU and D110/100. 2.) Error of GSSU 15VDC power supply; circuit breaker F2 in the GPA is switched off; low line voltage. 3.) Check for monitoring error.	None.
Error	20	GXXmsg.mc	Gradient Power Amplifier error: Hardware error (modulator defect). Please call Siemens Service	The modulator board D40 is defective.	1.a) Check 5V and +/-15V supply, green LEDs D40-V121 are on? 1.b) Is there also a controlling error? If yes, also look there. 1.c) Check jumper settings of D40. 1.d) Check cables from D40 to power stages for short circuit. 1.e) Check power stages. 2.) Check for monitoring error. 3.) Check also for accompanying error messages.	None.
Error	21	GXXmsg.mc	Gradient Power Amplifier error: GPA has been switched off by the Stimulation-Monitor. Check sequence parameter, reboot Scanner. If the problem still persists, please call Siemens Service	The stimulation monitor has switched off the GPA. Refer to the Stimulation Popup windows and follow the instructions given there.	Check, if stimulation limits have been exceeded. Check for other error messages from the GPA.	None.
Error	22	GXXmsg.mc	Gradient Power Amplifier error: An open COIL switch is detected during scan or scan prepare. Re-connect COIL. If the problem still persists, please call Siemens Service	The switching between different coils was not correct. The AMC want to start a sequence while the new coil was not in the loop.	Check coil switch.	none
Error	23	GXXmsg.mc	Gradient Power Amplifier error: Main power supply is defect. Please call Siemens Service	The main power supply for the GPA (primary voltage of transformer T1) is switched off by contactor K2.	1.a) Check the cable between X3/D70 and X1/D110. 1.b) Check the cables between D110 and D100. 1.c) Check D110 and D40/D70. 1.d) Check small signal power supply slot 4 (24V).	None.
Error	24	GXXmsg.mc	Gradient Power Amplifier error: Internal error (synchronization fault). Reboot Scanner. If the problem still persists, please call Siemens Service	The GPA is not synchronized with the system clock.	1.a) Check gradient data cable. 1.b) Check switch positions on D40 / S2-1..8 = off on off off on off off, see jumper list. 1.c) Check D10. 1.d) Check D40. 2.) Check for monitoring error.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Warning	25	GXXmsg.mc	Gradient Power Amplifier warning: Internal error (synchronization fault/service mode). Reboot Scanner. If the problem still persists, please call Siemens Service	The GPA is in service mode. The GPA is not synchronized with the system clock.	1.a) Check gradient data cable. 1.b) Check switch positions on D40 / S2-1..8 = off on off off on off off, see jumper list. 1.c) Check D10. 1.d) Check D40. 2.) Check for monitoring error.	None.
Error	26	GXXmsg.mc	Gradient Power Amplifier error: Hardware error (GSSU power supply). Please call Siemens Service	The GSSU power supply unit is defective. All 3 driver voltages of the power stages are out of tolerance	1.a) Check fuse F2 in GPA 1.b) Measure and adjust the output voltage at the output of slot 1 of the power supply (17.5V). 1.c) Check cable connection between E4A and X1/D110 and the power stages X2/D21; E4A to T11 2.) Check for monitoring error.	None.
Error	27	GXXmsg.mc	Gradient Power Amplifier error: The modulator of the GPA made a wrong controlling signal. Please call Siemens Service	The modulator of the GPA made a wrong controlling signal, danger for the power stages.	1.) Replace defective D40. Check jumper settings of a newly replaced D40. In rare cases a power stage or a cable from a power stage to the D40 may be defective. 2.) Check for monitoring error.	None.
Error	28	GXXmsg.mc	Gradient Power Amplifier error: Hardware error (cable pull protection). Please call Siemens Service.	Some cable connections between modulator and power stages are missing or defective.	1.a) Check cables/connectors between modulator and power stages. X5, X4, X3/D40 to X1/D21 of respective E6 (SCSI cable)	None.
Error	29	GXXmsg.mc	Gradient Power Amplifier error: Temperature of the gradient coil is too high. Check sequence parameter, retry in 10 minutes. If the problem still persists, please call Siemens Service	The gradient coil temperature is too high.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2.a) Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances.2.b) If temperature of gradient coil is ok, check output A4120 SUPF/U9; if light is of, replace A4120SUPF. 2.c) Check FOC from the patient table electronics A4120 SUPF/U9 to U1/D70 (light off = error). Look for kinks and dents. 2.d) Remove the FOC from U1/D70, let a strong pocket flashlight shine into U1 and press S1 of D70 (power on). If now while the pocket flashlight still is on the error is removed, then the GPA is o.k.	None.
Warning	30	GXXmsg.mc	Gradient Power Amplifier warning: Temperature of air too high. The system keeps working, but wait approximately 10 minutes before you start the next sequence. If the problem still persists, please call Siemens Service	The temperature of the GPA is too high.	1.) Check the ACS cooling system. 2.) Check for monitoring error, temperature is measured of D70 in the GSSU.	None.
Error	31	GXXmsg.mc	Gradient Power Amplifier error: Temperature of air too high. Check sequence parameter, reboot Scanner. If the problem still persists, please call Siemens Service	The temperature of the GPA is too high.	1.) Check the ACS cooling system. 2.) Check for monitoring error, temperature is measured of D70 in the GSSU.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	32	GXXmsg.mc	Gradient Power Amplifier error: Temperature of main power supply is too high. Retry after 10 minutes. If the problem still persists, please call Siemens Service	The main supply transformer T1 is in overtemperature condition. Please check the cabinet cooling system (ACS and fans).	1.a) Check sequence parameters and measurement conditions. 1.b) Check the ACS cooling system and the fans inside the GPA cabinet. 2.) Check for monitoring errors (defective temperature sensors of the transformer, D110, cable D110 to D70, D70) connections);	None.
Error	33	GXXmsg.mc	Gradient Power Amplifier error: No reaction of modulator. Retry, reboot Scanner. If the problem still persists, please call Siemens Service	The modulators cannot be switched into the d	1.a) Check modulator switches on D70 ('ON' position). 1.b) If they are in the 'ON' position, check LED status on D40; get rid of all errors, which could disable the modulator. 1.c) Run TestTools/ AMC/ PCI-TX and Gradient/GPA Comm/Status. 1.d) Check D40. 1.e) Check D70 and/or D60.	None.
Warning	88	GXXmsg.mc	Gradient Power Amplifier warning: Init state not reached. Regulator, Grad Curr and Slewrate values are missing (in service mode). Reboot Scanner. If the problem still persists, please call Siemens Service.	The GPA needs from the AMC the correct specific values for gradient current, slewrate and regulator settings for going to Power ON.	Check CAN-connection, do Regulator Adjustment.	None.
Error	34	GXXmsg.mc	Gradient Power Amplifier error: Init state not reached. Regulator, Grad Curr and Slewrate values are missing. Restart Scanner If the problem still persists, please call Siemens Service.	The GPA needs from the AMC the correct specific values for gradient current, slewrate and regulator settings for going to Power ON.	Check CAN-connection, do Regulator Adjustment	None.
Error	35	GXXmsg.mc	Gradient Power Amplifier error: The Gradient Power Amplifier has detected an oscillation. Please check the sequence parameters an start the sequence again. If the problem still persists, please call Siemens Service.	The GPA detected an oscillation of at least one Gradient axis.	To locate the axis disable each time the other 2 modulators (switch D70, or CAN command). Service mode extends the investigation time. Possible reasons for this error: - Wrong regulator setting, do regulator adjust - The customer made his own sequence, the sequence does not work as it should. - Defective gradient data generation, defective gradient data cable, defective DAC board, - do the DAC loop test. - Short circuit of the gradient coil. - Any defect within the regulation loop	None.
Error	36	GXXmsg.mc	Gradient Power Amplifier error: Internal error (invalid message code)/spare 2. Please call Siemens Service.	This message code is currently not used.	Replace D70, D40, D60.	None.
Error	37	GXXmsg.mc	Gradient Power Amplifier error: Power Stage defect. Please call Siemens Service.	The Power Pack D110 doesn't switch on because Power Stage is shorten.	Change the defective Power Stage E6	None.
Error	38	GXXmsg.mc	Gradient Power Amplifier error: Fan Error. Please call Siemens Service.	One of the GPA internal fans doesn't work.	1.) Wait for cool down; check the air condition system. 2.) Replace defective Fan assembly. 3.) Check for monitoring error (D110 - D70 - D40).	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	39	GXXmsg.mc	Gradient Power Amplifier error: Regulation error on X-axis. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The Regulation error on X-Axis too big (Gradient pulse edge too steep) or the regulation loop is open (e.g. no coil connected, modulator disabled) or instable mains or low input voltage.	1.a) Look for other GPA error messages. 1.b) Check sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.c) Run TestTools/AMC test and Gradient test for DAC. Check gradient data cable and connectors. Check waveforms with the gradient loop or with a scope. 1.d) Check the continuity of the load circuit. Check for loose cable connections. 1.e) Perform Regulator adjust/check. Check for other error messages. 1.f) Perform QA/Grad. Rise Time Check. 1.g) Check the fifteen 3-phase circuit breakers between outputs of T1 and the inputs of D100. 1.h) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13): 80 VDC < U < 450 VDC; all five voltages should be approximately the same. 1.i) Check current sensor and/or the connection between D22 and current sensors. Measure the current actual value at D22, X11 with scope -> No overshoots after the pulse ramps. 1.k) Measure the regulator output signal at D22/X61 with scope -> Voltage pulse < 10V. 1.l) Swap/Replace power stage.	None.
Warning	40	GXXmsg.mc	Gradient Power Amplifier warning: Regulation error on X-axis (in service mode). Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The Regulation error on X-Axis too big (Gradient pulse edge too steep) or the regulation loop is open (e.g. no coil connected, modulator disabled) or instable mains or low input voltage.	1.a) Look for other GPA error messages 1.b) Check sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.c) Run TestTools/ AMC test, run Gradient test for DAC. Check gradient data cable and connectors. Check waveforms with the gradient loop or with a scope. 1.d) Check the continuity of the load circuit. Check for loose cable connections. 1.e) Perform Regulator adjust/check. Check for other error messages. 1.f) Perform QA/Grad. Rise Time Check. 1.g) Check the fifteen 3-phase circuit breakers between outputs of T1 and the inputs of D100. 1.h) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13): 80 VDC < U < 450 VDC; all five voltages should be approximately the same. 1.i) Check current sensor and/or the connection between D22 and current sensors. Measure the current actual value at D22, X11 with scope -> No overshoots after the pulse ramps. 1.k) Measure the regulator output signal at D22/X61 with scope -> Voltage pulse < 10V. 1.l) Swap/Replace power stage.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	41	GXXmsg.mc	Gradient Power Amplifier error: Internal control error (current value error on X-axis). Please call Siemens Service.	The measured value of the actual gradient current on X-axis is not plausible.	1.a) Check supply/signal line from SENS to D22/X10 to E7 for continuity and shorts. 1.b) Check the gradient coil circuit and gradient filter assembly for ground shorts and sparks. 1.c) Replace D22. 1.d) Check the cable connections. 1.e) Measure the sensor signals at D22, X11 for X-axis. 1.f) Replace defective pair of current sensors (LA305S_SP1 and 867-700SM) (SENS). 2.) Check for monitoring error.	None.
Error	42	GXXmsg.mc	Gradient Power Amplifier error: Overcurrent on X-axis. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The actual current on X-axis exceeds the limit (calculation on D22).	1.a) Check the sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.b) Run AMC test. Check gradient data cable. Run Gradient test for DAC. 1.c) Measure the resistance of the load circuit, short if only a few mOhm, measure the line voltage. 1.d) Run the Regulator Adjust/Check. Replace defective D22. 1.e) Replace defective danfysik current sensor 867-700SM. 2.) Check for monitoring error.	None.
Error	43	GXXmsg.mc	Gradient Power Amplifier error: Current duty cycle for power stage X too high. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The duty cycle for X of the actual current is too high.	1.a) Check the sequence parameters. Check the system by running a high duty cycle sequence. 1.b) Run AMC test. Check gradient data cable. Run gradient test for DAC. 1.c) Run Regulator Adjust/Check. Replace defective D22. 2.) Check for monitoring error.	None.
Error	44	GXXmsg.mc	Gradient Power Amplifier error: Controlling frequency error, power stage X. Reboot Scanner. If the problem still persists, please call Siemens Service	The frequency of the control signals for the power stage X is not ok.	1.a) Reboot Scanner. (Reloading of regulator parameters). 1.b) Perform TestTools/AMC. 1.c) Perform the DAC loop test. 1.d) Perform Regulator Adjust. 2.) Check for monitoring error.	None.
Error	45	GXXmsg.mc	Gradient Power Amplifier error: Temperature of power stage X too high. Retry after 10 minutes. If the problem still persists, please call Siemens Service	The X power stage transistors are too hot.	1.a) Check the sequence parameters. Run an approved high duty cycle sequence. 1.b) Check the cooling system. 1.c) Check the cables between power stages and D40. 1.d) Replace defective D40 and/or power stage. 2.) Check for monitoring error.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	46	GXXmsg.mc	Gradient Power Amplifier error: Overvoltage on Power stage X. Retry, reboot Scanner. If the problem still persists, please call Siemens Service.	Line overvoltage or negative feeding for the power stage X by the load circuit.	1.a) Check line voltage. If necessary, set transformer according to the line voltage (especially if errors occur during start up). 1.b) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13) with enabled/disabled modulator: 80 VDC < U < 450 VDC; all five voltages should be approximately the same. If voltage not ok with disabled modulator -> Failure of mains voltage, transformer T1, supervision D100, cabling. If voltage not ok with enabled modulator -> Failure of D40, cabling D40 to power stage, power stage. 1.c) Replace defective D100. Check cabling X1/D100 to X8/D110 and X1/D110 to X3/D70, D70. 1.d) Replace defective power stage. 1.e) Check cable connection between modulator D40 and power stage. 1.f) Replace defective D40.	None.
Warning	47	GXXmsg.mc	Gradient Power Amplifier warning: Under voltage of power stage X. Check sequence parameter, retry. If the problem still persists, please call Siemens Service.	Occurrence of under voltage for X-axis was detected.	Check 100A curcuit breaker. Check secondary side automatic circuit breaker. Check the rectifier D100. Check jumper setting of D40. Signal flow: D100 - D110 - D70 - D40	none
Error	48	GXXmsg.mc	Gradient Power Amplifier error: Under voltage of power stage X. Check sequence parameter, retry. If the problem still persists, please call Siemens Service.	Occurrence of under voltage for X-axis was detected.	Check 100A curcuit breaker. Check secondary side automatic circuit breaker. Check the rectifier D100. Signal flow: D100 - D110 - D70 - D40	none
Error	49	GXXmsg.mc	Gradient Power Amplifier error: Driver power supply X defect. Please call Siemens Service.	The driver power supply for X power stage is defective.	1.a) Check the cable connections between GSSU power supply E4, D110 and power stage. Measure GSSU power supply E4 Slot 1: 17.5V. 1.b) Check the green LED's on D21/power stage; replace defective power stage. 2.) Check for monitoring error (D40).	None.
Error	50	GXXmsg.mc	Gradient Power Amplifier error: Current duty cycle error (Ohm power loss X). Please call Siemens Service.	Ohm power loss in X-axis.	1.a) Decrease the rating of the system by gradient pulses. 1.b) Check for high shim offset currents of the GPA, sequence parameters. 2.) Check for monitoring error (D22).	None.
Error	51	GXXmsg.mc	Gradient Power Amplifier error: Rectifier temp of X too high. Check sequence parameter, retry. Please call Siemens Service	The temperature of the rectifier assembly on D100 is too high.	1.a) Check the ACS cooling system, check water hoses to D100. 1.b) Check flat cable between D110 and D100. Replace defective rectifier board D100; for further checks refer to TSG of GPA. 2.) Check for monitoring error.	None.
Error	52	GXXmsg.mc	Gradient Power Amplifier error: Inductor temp of X too high. Check sequence parameter, retry. Please call Siemens Service	The regulator is oscillating or the fan doesn't run.	Check Regulator settings (Adjustment). Check for Fan Error.	
Error	53	GXXmsg.mc	Gradient Power Amplifier error: Power stage cable X defect. Please call Siemens Service	The cable between D40 and power stage is defect or pulled out.	Check connection between D40 (Modulator) and X-power stage E6. Check jumper setting of D40.	

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	54	GXXmsg.mc	Gradient Power Amplifier error: The measurement of the power stage voltage X was interrupted. Check whether the line voltage was out of tolerance, check the sequence for extreme power demand for the GPA, restart. If the problem still persists, please call Siemens Service.	There was no voltage feed back signal from the X power stage to the modulator for at least 1 second. Possible reasons: Power stage voltage >450V or <250V during at least 1 second	Investigate line voltage errors, broken fuses, extreme power conditions for the GPA. Replace defective D40, cable from power stage X to D40, power stage X, cable from power stage X to D100, D100	None.
Error	55	GXXmsg.mc	Gradient Power Amplifier error: Regulation error on Y-axis. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The Regulation error on Y-Axis too big (Gradient pulse edge too steep) or the regulation loop is open (e.g. no coil connected, modulator disabled...) or instable mains or low input voltage.	1.a) Look for other GPA error messages 1.b) Check sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.c) Run TestTools/AMC test, run Gradient test for DAC. Check gradient data cable and connectors. Check waveforms with the gradient loop or with a scope. 1.d) Check the continuity of the load circuit. Check for loose cable connections. 1.e) Perform Regulator adjust/check. Check for other error messages. 1.f) Perform QA/Grad. Rise Time Check. 1.g) Check the fifteen 3-phase circuit breakers between outputs of T1 and the inputs of D100. 1.h) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13): 80 VDC < U < 450 VDC; all five voltages should be approximately the same. 1.i) Check current sensor and/or the connection between D22 and current sensors. Measure the current actual value at D22, X12 with scope -> No overshoots after the pulse ramps. 1.k) Measure the regulator output signal at D22/X71 with scope -> Voltage pulse < 10V. 1.l) Swap/Replace power stage.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Warning	56	GXXmsg.mc	Gradient Power Amplifier warning: Regulation error on Y-axis (in service mode). Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The GPA is in service mode. The Regulation error on Y-Axis too big (Gradient pulse edge too steep) or the regulation loop is open (e.g. no coil connected, modulator disabled...) or instable mains or low input voltage.	1.a) Look for other GPA error messages 1.b) Check sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.c) Run TestTools/AMC test, run Gradient test for DAC. Check gradient data cable and connectors. Check waveforms with the gradient loop or with a scope. 1.d) Check the continuity of the load circuit. Check for loose cable connections. 1.e) Perform Regulator adjust/check. Check for other error messages. 1.f) Perform QA/Grad. Rise Time Check. 1.g) Check the fifteen 3-phase circuit breakers between outputs of T1 and the inputs of D100. 1.h) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13); 80 VDC < U < 450 VDC; all five voltages should be approximately the same. 1.i) Check current sensor and/or the connection between D22 and current sensors. Measure the current actual value at D22, X12 with scope -> No overshoots after the pulse ramps. 1.k) Measure the regulator output signal at D22/X71 with scope -> Voltage pulse < 10V. 1.l) Swap/Replace power stage.	None.
Error	57	GXXmsg.mc	Gradient Power Amplifier error: Internal control error (current value error on Y-axis). Please call Siemens Service.	The measured value of the actual gradient current on Y-axis is not plausible.	1.a) Check supply/signal line from SENS to D22/X10 for continuity and shorts. 1.b) Check the gradient coil circuit and gradient filter assembly for ground shorts and sparks. 1.c) Replace D22. 1.d) Check the cable connections. 1.e) Measure the sensor signals at D22, X12 for Y-axis. 1.f) Replace defective pair of current sensors (LA305S_SP1 and 867-700SM) (SENS). 2.) Check for monitoring error.	None.
Error	58	GXXmsg.mc	Gradient Power Amplifier error: Overcurrent on Y-axis. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The actual current on Y-axis exceeds the limit (calculation on D22).	1.a) Check the sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.b) Run AMC test. Check gradient data cable. Run Gradient test for DAC. 1.c) Measure the resistance of the load circuit, short if only a few mOhm, measure the line voltage. 1.d) Run the Regulator Adjust/Check. Replace defective D22. 1.e) Replace defective danfysik current sensor 867-700SM. 2.) Check for monitoring error.	None.
Error	59	GXXmsg.mc	Gradient Power Amplifier error: Current duty cycle for power stage Y too high. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The duty cycle for Y of the actual current is too high.	1.a) Check the sequence parameters. Check the system by running a high duty cycle sequence. 1.b) Run AMC test. Check gradient data cable. Run gradient test for DAC. 1.c) Run Regulator Adjust/Check. 2.) Check for monitoring error.	None.
Error	60	GXXmsg.mc	Gradient Power Amplifier error: Controlling frequency error, power stage Y. Reboot Scanner. If the problem still persists, please call Siemens Service	The frequency of the control signals for the power stage Y is not ok.	1.a) Reboot Scanner. (Reloading of regulator parameters). 1.b) Perform TestTools/AMC. 1.c) Perform the DAC loop test. 1.d) Perform Regulator Adjust. 2.) Check for monitoring error.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	61	GXXmsg.mc	Gradient Power Amplifier error: Temperature of power stage Y too high. Retry after 10 minutes. If the problem still persists, please call Siemens Service	The Y power stage transistors are too hot.	1.a) Check the sequence parameters. Run an approved high duty cycle sequence. 1.b) Check the cooling system. 1.c) Check the cables between power stages and D40. 1.d) Replace defective D40 and/or power stage. 2.) Check for monitoring error.	None.
Error	62	GXXmsg.mc	Gradient Power Amplifier error: Overvoltage on Power stage Y. Retry, reboot Scanner. If the problem still persists, please call Siemens Service.	Line overvoltage or negative feeding for the power stage Y by the load circuit.	1.a) Check line voltage. If necessary, set transformer according to the line voltage (especially if errors occur during start up). 1.b) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13) with enabled/disabled modulator: 80 VDC < U < 450 VDC; all five voltages should be approximately the same. If voltage not ok with disabled modulator -> Failure of mains voltage, transformer T1, supervision D100, cabling. If voltage not ok with enabled modulator -> Failure of D40, cabling D40 to power stage, power stage. 1.c) Replace defective D100. Check cabling X1/D100 to X8/D110 and X1/D110 to X3/D70, D70. 1.d) Replace defective power stage. 1.e) Check cable connection between modulator D40 and power stage. 1.f) Replace defective D40.	None.
Warning	63	GXXmsg.mc	Gradient Power Amplifier warning: Under voltage of power stage Y. Check sequence parameter, retry. If the problem still persists, please call Siemens Service.	Occurrence of under voltage for Y-axis was detected.	Check 100A curcuit breaker. Check secondary side automatic circuit breaker. Check the rectifier D100. Check jumper setting of D40. Signal flow: D100 - D110 - D70 - D40	none
Error	64	GXXmsg.mc	Gradient Power Amplifier error: Under voltage of power stage Y. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	Occurrence of under voltage for Y-axis was detected.	Check 100A curcuit breaker. Check secondary side automatic circuit breaker. Check the rectifier D100. Signal flow: D100 - D110 - D70 - D40	none
Error	65	GXXmsg.mc	Gradient Power Amplifier error: Driver power supply Y defect. Please call Siemens Service.	The driver power supply for Y power stage is defective.	1.a) Check the cable connections between GSSU power supply, D110 and power stage. Measure GSSU power supply E4 Slot 1: 17.5V. 1.b) Check the green LED's on D21/power stage; 2.) Check for monitoring error (D40).	None.
Error	67	GXXmsg.mc	Gradient Power Amplifier error: Rectifier temp of Y too high. Check sequence parameter, retry. Please call Siemens Service	The temperature of the rectifier assembly on D100 is too high.	1.a) Check the ACS cooling system, check water hoses to D100. 1.b) Check flat cable between D110 and D100. Replace defective rectifier board D100; for further checks refer to TSG of GPA. 2.) Check for monitoring error.	None.
Error	66	GXXmsg.mc	Gradient Power Amplifier error: Current duty cycle error (Ohm power loss Y). Please call Siemens Service.	Ohm power loss in Y-axis.	1.a) Decrease the rating of the system by gradient pulses. 1.b) Check for high shim offset currents of the GPA, sequence parameters. 2.) Check for monitoring error (D22).	None.
Error	68	GXXmsg.mc	Gradient Power Amplifier error: Inductor temp of Y too high. Check sequence parameter, retry. Please call Siemens Service	The regulator is oscillating or the fan doesn't run.	Check Regulator settings (Adjustment). Check for Fan Error.	
Error	69	GXXmsg.mc	Gradient Power Amplifier error: Power stage cable Y defect Please call Siemens Service	The cable between D40 and power stage is defect or pulled out.	Check connection between D40 (Modulator) and Y-power stage E6 Check jumper setting of D40.	

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	70	GXXmsg.mc	Gradient Power Amplifier error: The measurement of the power stage voltage Y was interrupted. Check whether the line voltage was out of tolerance, check the sequence for extreme power demand for the GPA, restart. If the problem still persists, please call Siemens Service.	There was no voltage feed back signal from the Y power stage to the modulator for at least 1 second. Possible reasons: Power stage voltage >450V or <250V during at least 1 second	Investigate line voltage errors, broken fuses, extreme power conditions for the GPA. Replace defective D40, cable from power stage Y to D40, power stage Y, cable from power stage Y to D100, D100	None.
Error	71	GXXmsg.mc	Gradient Power Amplifier error: Regulation error on Z-axis. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The Regulation error on Z-Axis too big (Gradient pulse edge too steep) or the regulation loop is open (e.g. no coil connected, modulator disabled...) or instable mains or low input voltage.	1.a) Look for other GPA error messages 1.b) Check sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.c) Run TestTools/AMC test, run Gradient test for DAC. Check gradient data cable and connectors. Check waveforms with the gradient loop or with a scope. 1.d) Check the continuity of the load circuit. Check for loose cable connections. 1.e) Perform Regulator adjust/check. Check for other error messages. 1.f) Perform QA/Grad. Rise Time Check. 1.g) Check the fifteen 3-phase circuit breakers between outputs of T1 and the inputs of D100. 1.h) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13): 80 VDC < U < 450 VDC; all five voltages should be approximately the same. 1.i) Check current sensor and/or the connection between D22 and current sensors. Measure the current actual value at D22, X13 with scope -> No overshoots after the pulse ramps. 1.k) Measure the regulator output signal at D22/X81 with scope -> Voltage pulse < 10V. 1.l) Swap/Replace power stage.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Warning	72	GXXmsg.mc	Gradient Power Amplifier warning: Regulation error on Z-axis (in service mode). Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The GPA is in service mode. The Regulation error on Z-Axis too big (Gradient pulse edge too steep) or the regulation loop is open (e.g. no coil connected, modulator disabled...) or instable mains or low input voltage.	1.a) Look for other GPA error messages 1.b) Check sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.c) Run TestTools/AMC test, run Gradient test for DAC. Check gradient data cable and connectors. Check waveforms with the gradient loop or with a scope. 1.d) Check the continuity of the load circuit. Check for loose cable connections. 1.e) Perform Regulator adjust/check. Check for other error messages. 1.f) Perform QA/Grad. Rise Time Check. 1.g) Check the fifteen 3-phase circuit breakers between outputs of T1 and the inputs of D100. 1.h) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13); 80 VDC < U < 450 VDC; all five voltages should be approximately the same. 1.i) Check current sensor and/or the connection between D22 and current sensors. Measure the current actual value at D22, X13 with scope -> No overshoots after the pulse ramps. 1.k) Measure the regulator output signal at D22/X81 with scope -> Voltage pulse < 10V. 1.l) Swap/Replace power stage.	None.
Error	73	GXXmsg.mc	Gradient Power Amplifier error: Internal control error (current value error on Z-axis). Please call Siemens Service.	The measured value of the actual gradient current on Z-axis is not plausible.	1.a) Check supply/signal line from SENS to D22/X10 for continuity and shorts. 1.b) Check the gradient coil circuit and gradient filter assembly for ground shorts and sparks. 1.c) Replace D22. 1.d) Check the cable connections. 1.e) Measure the sensor signals at D22, X13 for Z-axis. 1.f) Replace defective pair of current sensors (LA305S_SP1 and 867-700SM) (SENS). 2.) Check for monitoring error.	None.
Error	74	GXXmsg.mc	Gradient Power Amplifier error: Overcurrent on Z-axis. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The actual current on Z-axis exceeds the limit (calculation on D22).	1.a) Check the sequence parameters (Gradient pulse edge too steep, if other than SIEMENS released sequences are used). 1.b) Run AMC test. Check gradient data cable. Run Gradient test for DAC. 1.c) Measure the resistance of the load circuit, short if only a few mOhm, measure the line voltage. 1.d) Run the Regulator Adjust/Check.. 1.e) Replace defective danfysik current sensor 867-700SM. 2.) Check for monitoring error.	None.
Error	75	GXXmsg.mc	Gradient Power Amplifier error: Current duty cycle for power stage Z too high. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	The duty cycle for Z of the actual current is too high.	1.a) Check the sequence parameters. Check the system by running a high duty cycle sequence. 1.b) Run AMC test. Check gradient data cable. Run gradient test for DAC. 1.c) Run Regulator Adjust/Check. 2.) Check for monitoring error.	None.
Error	76	GXXmsg.mc	Gradient Power Amplifier error: Controlling frequency error, power stage Z. Reboot Scanner. If the problem still persists, please call Siemens Service	The frequency of the control signals for the power stage Z is not ok.	1.a) Reboot Scanner. (Reloading of regulator parameters). 1.b) Perform TestTools/AMC. 1.c) Perform the DAC loop test. 1.d) Perform Regulator Adjust. 2.) Check for monitoring error.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	77	GXXmsg.mc	Gradient Power Amplifier error: Temperature of power stage Z too high. Retry after 10 minutes. If the problem still persists, please call Siemens Service	The Z power stage transistors are too hot.	1.a) Check the sequence parameters. Run an approved high duty cycle sequence. 1.b) Check the cooling system. 1.c) Check the cables between power stages and D40. 1.d) Replace defective D40 and/or power stage. 2.) Check for monitoring error.	None.
Error	78	GXXmsg.mc	Gradient Power Amplifier error: Overvoltage on Power stage Z. Retry, reboot Scanner. If the problem still persists, please call Siemens Service.	Line overvoltage or negative feeding for the power stage Z by the load circuit.	1.a) Check line voltage. If necessary, set transformer according to the line voltage (especially if errors occur during start up). 1.b) Measure the output voltage of the main power supply at connected plug X31/X32 at power stage. X31(1-4, 7-10), X31(13)-X32(1), X32(4-7), X32(10-13) with enabled/disabled modulator: 80 VDC < U < 450 VDC; all five voltages should be approximately the same. If voltage not ok with disabled modulator -> Failure of mains voltage, transformer T1, supervision D100, cabling. If voltage not ok with enabled modulator -> Failure of D40, cabling D40 to power stage, power stage. 1.c) Replace defective D100. Check cabling X1/D100 to X8/D110 and X1/D110 to X3/D70, D70. 1.d) Replace defective power stage. 1.e) Check cable connection between modulator D40 and power stage. 1.f) Replace defective D40.	None.
Warning	79	GXXmsg.mc	Gradient Power Amplifier warning: Under voltage of power stage Z. Check sequence parameter, retry. If the problem still persists, please call Siemens Service	Occurence of under voltage for Z-axis was detected.	Check 100A curcuit breaker. Check secondary side automatic circuit breaker. Check the rectifier D100. Check jumper setting of D40. Signal flow: D100 - D110 - D70 - D40	none
Error	80	GXXmsg.mc	Gradient Power Amplifier error: Under voltage of power stage Z. Please call Siemens Service.	Occurence of under voltage for Z-axis was detected.	Check 100A curcuit breaker. Check secondary side automatic circuit breaker. Check the rectifier D100. Signal flow: D100 - D110 - D70 - D40	none
Error	81	GXXmsg.mc	Gradient Power Amplifier error: Driver power supply Z defect. Please call Siemens Service.	The driver power supply for Z power stage is defective.	1.a) Check the cable connections between GSSU power supply, D110 and power stage. Measure GSSU power supply E4 Slot 1: 17.5V. 1.b) Check the green LED's on D21/power stage; replace defective power stage. 2.) Check for monitoring error (D40).	None.
Error	82	GXXmsg.mc	Gradient Power Amplifier error: Current duty cycle error (Ohm power loss Z). Please call Siemens Service.	Ohm power loss in Z-axis.	1.a) Decrease the rating of the system by gradient pulses. 1.b) Check for high shim offset currents of the GPA, sequence arameters. 2.) Check for monitoring error (D22).	None.
Error	83	GXXmsg.mc	Gradient Power Amplifier error: Rectifier temp od Z too high. Check sequence parameter, retry. Please call Siemens Service	The temperature of the rectifier assembly on D100 is too high.	1.a) Check the ACS cooling system, check water hoses to D100. 1.b) Check flat cable between D110 and D100. Replace defective rectifier board D100;. 2.) Check for monitoring error.	None.
Error	84	GXXmsg.mc	Gradient Power Amplifier error: Inductor temp of Z too high. Check sequence parameter, retry. Please call Siemens Service	The regulator is oscillating or the fan doesn't run.	Check Regulator settings (Adjustment). Check for Fan Error.	
Error	85	GXXmsg.mc	Gradient Power Amplifier error: Power stage cable Z defect Please call Siemens Service	The cable between D40 and power stage is defect or pulled out.	Check connection between D40 (Modulator) and Z-power stage E6 Check jumper setting of D40.	

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	86	GXXmsg.mc	Gradient Power Amplifier error: The measurement of the power stage voltage Z was interrupted. Check whether the line voltage was out of tolerance, check the sequence for extreme power demand for the GPA, restart. If the problem still persists, please call Siemens Service.	There was no voltage feed back signal from the Z power stage to the modulator for at least 1 second. Possible reasons: Power stage voltage >450V or <250V during at least 1 second	Investigate line voltage errors, broken fuses, extreme power conditions for the GPA. Replace defective D40, cable from power stage Z to D40, power stage Z, cable from power stage Z to D100, D100	None.
Error	87	GXXmsg.mc	Gradient Power Amplifier error: The CPU IO pins could not set correct. Please call Siemens Service.	An internal GXXLW error occurred during accessing the GPA hardware	Check all small signal boards.	None.
Error	89	GXXmsg.mc	Gradient Power Amplifier error: The Loadware detect a switch to Pre Operational or Stopp State. The modulator of the GPA made a soft stop. Please call Siemens Service	The Loadware detect a switch to Pre Operational or Stopp State. The modulator of the GPA made a soft stop. Now the GXXLW will not monitor the GPA status further on.	Switch the GXXLW to the state OPERATIONAL again.	None.
Warning	91	GXXmsg.mc	Gradient Power Amplifier warning: The measurement of the power stage voltage X was interrupted, currently the GPA is in Service mode. The sequence was not interrupted, but please check whether the line voltage was out of tolerance, check the sequence for extreme power demand for the GPA, restart. If the problem still persists, please call Siemens Service.	There was no voltage feed back signal from the X power stage to the modulator for at least 1 second, while the GPA was in service mode. Possible reasons: Power stage voltage >450V or <250V during at least 1 second	Investigate line voltage errors, broken fuses, extreme power conditions for the GPA. Replace defective D40, cable from power stage X to D40, power stage X, cable from power stage X to D100, D100	None.
Warning	92	GXXmsg.mc	Gradient Power Amplifier warning: The measurement of the power stage voltage Y was interrupted, currently the GPA is in Service mode. The sequence was not interrupted, but please check whether the line voltage was out of tolerance, check the sequence for extreme power demand for the GPA, restart. If the problem still persists, please call Siemens Service.	There was no voltage feed back signal from the Y power stage to the modulator for at least 1 second, while the GPA was in service mode. Possible reasons: Power stage voltage >450V or <250V during at least 1 second	Investigate line voltage errors, broken fuses, extreme power conditions for the GPA. Replace defective D40, cable from power stage Y to D40, power stage Y, cable from power stage Y to D100, D100	None.
Warning	93	GXXmsg.mc	Gradient Power Amplifier warning: The measurement of the power stage voltage Z was interrupted, currently the GPA is in Service mode. The sequence was not interrupted, but please check whether the line voltage was out of tolerance, check the sequence for extreme power demand for the GPA, restart. If the problem still persists, please call Siemens Service.	There was no voltage feed back signal from the Y power stage to the modulator for at least 1 second, while the GPA was in service mode. Possible reasons: Power stage voltage >450V or <250V during at least 1 second	Investigate line voltage errors, broken fuses, extreme power conditions for the GPA. Replace defective D40, cable from power stage Z to D40, power stage Z, cable from power stage Z to D100, D100	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	2000	GXXmsg.mc	Gradient Power Amplifier error: The GPA is not ready for measurement. Retry, reboot Scanner, switch off/on Scanner. If the problem still persists, please call Siemens Service	None.	* Start Test Tools and check GPA status. Is the unit switched off or working in firmware context? * Evaluate previous error messages. * Reboot Scanner. * If not successful, switch off/on Scanner.	None.
Error	2001	GXXmsg.mc	Gradient Power Amplifier error: While scanning, the power supply for the GPA was switched off. Retry, reboot Scanner, switch off/on Scanner. If the problem still persists, please call Siemens Service	Power supply was switched off due to overtemperature or other fault.	* Start Test Tools and check GPA status. * Evaluate previous error messages. * Reboot Scanner. * If not successful, switch off/on Scanner.	None.
Error	2002	GXXmsg.mc	Gradient Power Amplifier error: Temperature of the gradient coil is too high. Retry after 10 minutes. If the problem still persists, please call Siemens Service	The gradient coil temperature is too high.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2.a) Check the signal lines between the gradient coil NTC's and D9. Turn off defective NTC's via DIP switch on D9; measure the NTC resistance. 2.b) Check cables and connectors on D9. 2.c) Check on D9 the red LED's V4,V12,V20 and the position of DIP switches. Replace defective D9. 2.d) Check FOC from U1/D9 to U7/D17 (light off = error).	None.
Error	2004	GXXmsg.mc	Gradient Power Amplifier error: Stimulation limit exceeded. Change the sequence parameter.	The sequence has been stopped because the stimulation monitor has detected that the stimulation limit has been exceeded. Please change the protocol/sequence parameters before restarting the measurement..	Please try at least one of the following recommendations: 1. Increase the FOV 2. Change the SWAP mode 3. Reduce the matrix size 4. Increase the slice thickness 5. Decrease the phase resolution/the number of lines	None.
Warning	3000	GXXmsg.mc	Gradient Power Amplifier warning: While scanning, the GPA left the "sleep" mode. This could cause noise in the images. Retry. If the problem still persists, please call Siemens Service	GPA CAN woke up while scanning.	* Start Test Tools and check the status of all CAN units. * Evaluate previous error messages. * Start sequence again. * If not successful, reboot Scanner to solve a possible software problem. * If not successful, reset hardware by switching off/on Scanner.	None.
Error	3001	GXXmsg.mc	Gradient Power Amplifier error: No communication with unit GPA. Retry, reboot Scanner, switch off/on Scanner. If the problem still persists, please call Siemens Service	Communication with GPA not possible. GPA communication timeout occurred.	* Start Test Tools and run CAN tests. * Evaluate previous error messages. * Reboot Scanner to solve a possible software hang-up. * If not successful, reset hardware by switching off/on Scanner.	None.
Error	3002	GXXmsg.mc	Gradient Power Amplifier error: GPA Device fault. Retry, reboot Scanner, switch off/on Scanner. If the problem still persists, please call Siemens Service	GPA device fault	* Start Test Tools and run CAN tests. * Evaluate previous error messages. * Reboot Scanner to solve a possible software hang-up. * If not successful, reset hardware by switching off/on Scanner.	None.

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Warning	3003	GXXmsg.mc	Gradient Power Amplifier warning: GPA not operational. Retry, reboot Scanner, switch off/on Scanner. If the problem still persists, please call Siemens Service	GPA not operational.	* Start Test Tools and run CAN tests. * Evaluate previous error messages. * Reboot Scanner to solve a possible software hang-up. * If not successful, reset hardware by switching off/on Scanner.	None.
Error	3004	GXXmsg.mc	Gradient Power Amplifier error: The GPA was reset unexpectedly. Retry, reboot Scanner, switch off/on Scanner. If the problem still persists, please call Siemens Service.	Logon message received due to a reset of GPA CAN.	* Start Test Tools and check the status of all CAN units. * Search previous error messages, particularly for temperature and power faults. * Reboot Scanner and start sequence again.	None.
Error	3005	GXXmsg.mc	Gradient Power Amplifier error: Configuration was not successful. Switch off/on Scanner. Please call Siemens Service.	Shim device fault	* Start Test Tools and run CAN tests. * Evaluate previous error messages. * Reboot Scanner to solve a possible software hang-up. * If not successful, reset hardware by switching off/on Scanner.	None.
Warning	5100	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "PRI_X" is hot. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5101	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "PRI_X" is too hot. Wait for cooling down for approximately 10 minutes! Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "PRI_X" is over the error threshold during scanner boot.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts" 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5102	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "PRI_Y" is hot. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5103	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "PRI_Y" is too hot. Wait for cooling down for approximately 10 minutes! Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "PRI_Y" is over the error threshold during scanner boot.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts" 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Warning	5104	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "PRI_Z" is hot. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5105	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "PRI_Z" is too hot. Wait for cooling down for approximately 10 minutes! Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "PRI_Z" is over the error threshold during scanner boot.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts" 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5106	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SEC_X" is hot. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5107	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SEC_X" is too hot. Wait for cooling down for approximately 10 minutes! Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SEC_X" is over the error threshold during scanner boot.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts" 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5108	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SEC_Y" is hot. Reboot Scanner! If the problem still persists, please call Siemens Service			

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Error	5109	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SEC_Y" is too hot. Wait for cooling down for approximately 10 minutes! Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SEC_Y" is over the error threshold during scanner boot.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts" 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5110	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SEC_Z" is hot. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5111	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SEC_Z" is too hot. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SEC_Z" is over the error threshold during scanner boot.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts" 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5112	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SOCKET_1" is hot. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5113	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SOCKET_1" is too hot. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SOCKET_1" is over the error threshold during scanner boot.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts" 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Warning	5114	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SOCKET_2" is hot. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5115	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SOCKET_2" is too hot. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SOCKET_2" is over the error threshold during scanner boot.	1.a) Wait (approx. 10 min.) for cooling down of the gradient coil and try again. Reduce the thermal load of the coil by appropriate measurement conditions. 1.b) Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts" 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5200	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "PRI_X" is cold. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5201	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "PRI_X" is too cold. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "PRI_X" is under the error threshold during scanner boot.	1. Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts". 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5202	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "PRI_Y" is cold. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5203	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "PRI_Y" is too cold. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "PRI_Y" is under the error threshold during scanner boot.	1. Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts". 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Warning	5204	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "PRI_Z" is cold. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5205	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "PRI_Z" is too cold. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "PRI_Z" is under the error threshold during scanner boot.	1. Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts". 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5206	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SEC_X" is cold. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5207	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SEC_X" is too cold. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SEC_X" is under the error threshold during scanner boot.	1. Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts". 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5208	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SEC_Y" is cold. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5209	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SEC_Y" is too cold. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SEC_Y" is under the error threshold during scanner boot.	1. Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts". 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	

Message class S/I/W/E	Id	Source	Message Text	Explanation	Action	Information
Warning	5210	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SEC_Z" is cold. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5211	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SEC_Z" is too cold. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SEC_Z" is under the error threshold during scanner boot.	1. Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts". 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5212	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SOCKET_1" is cold. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5213	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SOCKET_1" is too cold. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SOCKET_1" is under the error threshold during scanner boot.	1. Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts". 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	
Warning	5214	PXXmsg.mc	Patient Table warning: Gradient coil temperature sensor "SOCKET_2" is cold. Reboot Scanner! If the problem still persists, please call Siemens Service			
Error	5215	PXXmsg.mc	Patient Table error: Gradient coil temperature sensor "SOCKET_2" is too cold. Reboot Scanner! If the problem still persists, please call Siemens Service	Temperature value from gradient coil sensor "SOCKET_2" is under the error threshold during scanner boot.	1. Check the cooling system. 2. Reboot Scanner, if the problem still persists continue. 3. Check the signal lines between the gradient coil NTC's, X2 and the patient table electronics, A4120 SUPF/X50 (refer to wiring diagrams). Measure and compare the NTC resistances. 4. Install the spare sensors according to document "Replacement of Parts". 5. Reboot Scanner 6. If error occurs again, create Savelog and inform HSC for further investigation	